

Bingjie Tang

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RESEARCH INTERESTS

I am broadly interested in robotic manipulation, with a focus on learning sim-to-real transferable skills for contact-rich manipulation and developing tools for efficient and reliable sim-to-real transfer.

EDUCATION

- **University of Southern California** Aug. 2020 - May 2025
PhD in Computer Science, advised by Gaurav S. Sukhatme
Los Angeles, USA
- **Brown University** Aug. 2018 - May 2020
MS in Computer Science, advised by George D. Konidaris & Stefanie A. Tellex.
Providence, USA
- **Huazhong University of Science and Technology** Aug. 2014 - May 2018
BS in Computer Science
Wuhan, China

PATENTS AND PUBLICATIONS

- [1] **Bingjie Tang**, Matthew Corsaro, Gero Konidaris, Stefanos Nikolaidis, Stefanie Tellex. **Learning Collaborative Pushing and Grasping Policies in Dense Clutter**. *International Conference on Robotics & Automation (ICRA)*, 2021.
- [2] **Bingjie Tang**, Gaurav S. Sukhatme. **Selective Object Rearrangement in Clutter**. *6th Annual Conference on Robot Learning (CoRL)*, 2022.
- [3] **Bingjie Tang***, Michael Lin*, Iretey Akinola, Ankur Handa, Gaurav Sukhatme, Fabio Ramos, Dieter Fox, Yashraj Narang. **IndustReal: Transferring Contact-Rich Assembly Tasks from Simulation to Reality**. *19th Robotics: Science and Systems (RSS)*, 2023.
- [4] **Bingjie Tang**, Yashraj Narang, Dieter Fox, Fabio Ramos. **Techniques for training and implementing reinforcement learning policies for robot control**. *US Patent App. 18/489,789*, 2024.
- [5] **Bingjie Tang**, Iretey Akinola, Jie Xu, Bowen Wen, Ankur Handa, Karl Van Wyk, Dieter Fox, Gaurav Sukhatme, Fabio Ramos, Yashraj Narang. **AutoMate: Specialist and Generalist Assembly Policies over Diverse Geometries**. *20th Robotics: Science and Systems (RSS)*, 2024.
- [6] Yian Wang, **Bingjie Tang**, Chuang Gan, Dieter Fox, Kaichun Mo, Yashraj Narang, Iretey Akinola. **MatchMaker: Automated Robotic Assembly Asset Generation for Policy Learning in Simulation**. *International Conference on Robotics & Automation (ICRA)*, 2025.
- [7] Yijie Guo, **Bingjie Tang**, Iretey Akinola, Dieter Fox, Abhishek Gupta, Yashraj Narang. **SRSA: Skill Retrieval and Adaptation for Robotic Assembly Tasks**. *International Conference on Learning Representations (ICLR)*, 2025.
- [8] Michael Noseworthy, **Bingjie Tang**, Bowen Wen, Ankur Handa, Nicholas Roy, Dieter Fox, Fabio Ramos, Yashraj Narang, Iretey Akinola. **Forge: Force-guided exploration for robust contact-rich manipulation under uncertainty**. *IEEE Robotics and Automation Letters (RA-L)*.
- [9] Nicklas Hansen, Iretey Akinola, Yijie Guo, Jie Xu, **Bingjie Tang**, Hao Su, Xiaolong Wang, Abhishek Gupta, Dieter Fox, Yashraj Narang. **Generalizable Robotic Insertion with World Models** *In submission*.
- [10] **Bingjie Tang**, Iretey Akinola, Jie Xu, Bowen Wen, Michael Andres Lin, Dieter Fox, Gaurav S. Sukhatme, Fabio Ramos, Abhishek Gupta, Yashraj Narang. **Refinery: Active Fine-tuning and Deployment-time Optimization for Contact-Rich Policies** *International Conference on Robotics & Automation (ICRA)*, 2026.
- [11] Nicklas Hansen, Iretey Akinola, Yijie Guo, Jie Xu, **Bingjie Tang**, Hao Su, Xiaolong Wang, Abhishek Gupta, Dieter Fox, Yashraj Narang. **Generalizable Robotic Insertion with World Models** *International Conference on Intelligent Robots and Systems (IROS)*, 2026.

WORKING EXPERIENCE

- **Optimus AI, Tesla** July 2025 - Present
Senior Robotics Machine Learning Engineer
Palo Alto, USA
- **Seattle Robotics Lab, NVIDIA Corporation** [🌐] May 2022 - May 2025
Robotics Research Intern, advised by Yashraj Narang & Dieter Fox
Seattle, USA
- **Technology Engineering Group (TEG), Tencent** Jun. 2017 - Sept. 2017
Software Engineer Intern
Shenzhen, China

TEACHING EXPERIENCE

- **CSCI 1951-R: Introduction to Robotics** *Brown University*
Instructor: Stefanie Tellex 2019 Fall
- **CSCI 566: Deep Learning and its Applications** *University of Southern California*
Instructor: Xiang Ren/Jesse Thomason/Yan Liu 2022/2023 Spring, 2024 Fall
- **CSCI 455: Introduction to Robotics** *University of Southern California*
Instructor: Heather Culbertson 2022 Fall
- **CSCI 677: Advanced Computer Vision** *University of Southern California*
Instructor: Yue Wang 2023 Fall

PROFESSIONAL SERVICE

- **Associate editor:** ICRA 2026.
- **Reviewer:** ICRA, IROS, CoRL, RSS, RA-L, T-RO.

SKILLS

- **Programming Languages:** Python, C, C++.
- **Deep Learning Libraries:** Pytorch, Tensorflow.
- **Robotics:** ROS, Pybullet, CoppeliaSim, IsaacGym, IsaacLab